

Kubernetes Security

Since you're learning **KCNA as a beginner**, let's understand Kubernetes Security step by step.

1. What is Kubernetes Security?

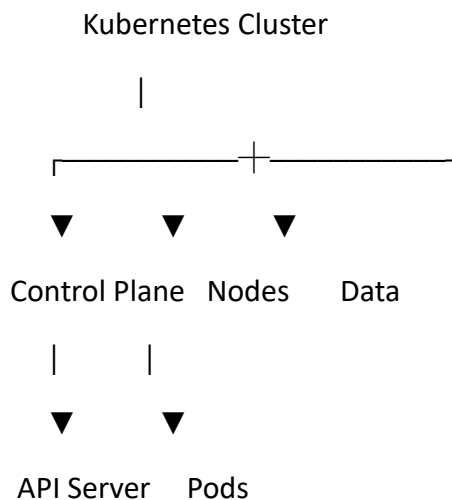
Kubernetes Security means protecting the Kubernetes cluster, its Nodes, Pods, applications, and data from unauthorized access or harmful activity.

In simple words:

 **Kubernetes Security = Making sure only the right people and applications can access the right things.**

2. What Needs to Be Protected?

A Kubernetes cluster has many parts:



We need to protect:

-  **API Server**
-  **Control Plane**
-  **Worker Nodes**
-  **Pods**
-  **Containers**
-  **Secrets**

-  **Network communication**
 -  **Applications and data**
-

3. Main Kubernetes Security Areas

For KCNA, remember these major areas:

Kubernetes Security

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├— Authentication

├— Authorization

├— Admission Control

├— Secrets

├— RBAC

├— Network Policies

└— Pod Security

Let's understand each one.

4. Authentication

What is Authentication?

Authentication checks WHO you are.

Example:

You want to access Kubernetes.

Kubernetes asks:

Who are you?

You provide credentials.

For example:

Username / Certificate / Token

If your identity is valid:

✅ You are authenticated.

Easy Memory

👤 **Authentication = WHO are you?**

5. Authorization 📖

After Kubernetes knows who you are, it asks:

What are you allowed to do?

For example:

User: Ali

Can Ali:

✓ View Pods?

✓ Create Pods?

✗ Delete Nodes?

Authorization determines the permissions.

Easy Memory

🔑 **Authorization = WHAT are you allowed to do?**

6. Authentication vs Authorization ★

This is very important.

Authentication	Authorization
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Who are you?	What can you do?
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Verifies identity	Verifies permissions
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Authentication Authorization

Login/credentials Access rules

Easy Trick

Authentication → WHO? 

Authorization → WHAT? 

7. RBAC

RBAC means:

Role-Based Access Control

It controls what users or service accounts are allowed to do based on their assigned roles.

Example:

Developer



Can view Pods

Can view Deployments

 Cannot delete Nodes

Another user:

Administrator



More permissions

Easy Memory

 **RBAC = Permissions based on Roles**

8. Kubernetes Secrets

Secrets are used to store sensitive information.

Examples:

- Passwords
- API keys
- Tokens
- Certificates

Example:

Database Password



Secret 



Pod

Instead of putting a password directly inside your application configuration, you can use a Kubernetes Secret.

Easy Memory

 **Secret = Sensitive information**

9. Network Policies

Network Policies control **which Pods can communicate with which other Pods.**

Imagine:

Frontend Pod



Backend Pod

You may want:

Frontend → Backend 

Frontend → Database ❌

A NetworkPolicy can help control this traffic.

Easy Memory

 **NetworkPolicy = Controls Pod-to-Pod network traffic**

10. Pod Security

Pods and containers should be configured securely.

For example, you may want to prevent containers from having unnecessary privileges.

Good security practices include:

- Don't give containers unnecessary privileges.
- Use appropriate user permissions.
- Avoid running applications as root when not required.
- Use trusted container images.
- Keep images updated.

Easy Memory

 **Pod Security = Make Pods/containers safer**

11. API Server Security

Remember:

API Server is the front door of Kubernetes.

Almost everything communicates through the API Server.

Therefore, it must be protected.

User

|



 API Server

|



Kubernetes Cluster

Security mechanisms such as:

- Authentication
- Authorization
- Admission control

help protect access to the API.

12. Admission Control

After authentication and authorization, Kubernetes can use **admission control** to inspect or modify requests before they are accepted.

Simple flow:

User

|



Authentication

|



Authorization

|



Admission Control

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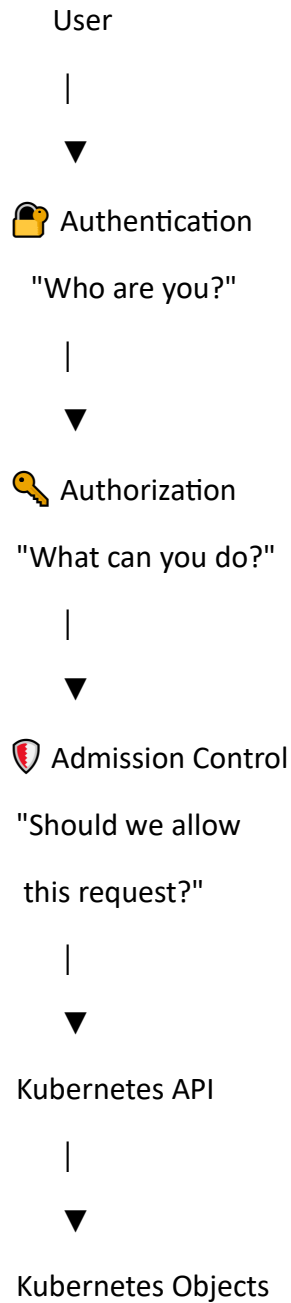


API Server accepts request

Think of admission control as a **security checkpoint**.

13. Complete Security Flow ★

This is a useful diagram to remember:



14. Real-Life Example

Imagine entering an office.

Step 1 — Authentication

Security guard asks:

"Who are you?"

You show your ID.

 Identity verified.

Step 2 — Authorization

Guard checks:

"Are you allowed to enter this room?"

Maybe:

Employee → Office 

Visitor → Server Room 

Step 3 — Admission Control

There may be additional security rules:

"Does this visitor have an appointment?"

If yes:

 Allow.

If no:

 Reject.

This is similar to Kubernetes security.

15. Important KCNA Revision Table

Topic	Easy Meaning
Authentication	Who are you?
Authorization	What can you do?
RBAC	Permissions based on roles
Secrets	Store sensitive information
NetworkPolicy	Controls Pod network traffic
Pod Security	Protect Pods/containers
Admission Control	Checks requests before acceptance
API Server Security	Protects Kubernetes entry point

Super Easy Memory Trick

Remember:

WHO? → Authentication 

WHAT? → Authorization 

ROLE? → RBAC 

PASSWORD? → Secret 

NETWORK? → NetworkPolicy 

POD? → Pod Security 

REQUEST? → Admission Control 

★ **One-line KCNA definition**

Kubernetes Security protects the cluster and its resources by controlling identity, permissions, network communication, sensitive data, and Pod behavior.